

```
align(center + horizon)[ block( width: 15cm, inset: (y: 1.1cm), stroke: (top: 2.5pt + rgb(26, 54, 93),
bottom: 0.75pt + rgb(203, 213, 225)), ) [ align(center) [ text(size: 10.5pt, tracking: 0.35em, fill:
rgb(71, 85, 105)) [大 作 业] v(0.55cm) text(size: 26pt, weight: "bold", fill: rgb(15, 23, 42)) [揭棋对弈
程序设计] v(0.65cm) text(size: 19pt, weight: "medium", fill: rgb(30, 64, 175)) [构建与运行]
v(0.35cm) if subtitle != "" [ text(size: 13pt, fill: rgb(100, 116, 139)) [Build & Run Guide — 环境、构
建、演示流程] v(0.35cm) ] ] ]
```

```
v(1.5cm)
```

```
box( width: 13cm, inset: (x: 1.2cm, y: 0.95cm), fill: rgb(248, 250, 252), radius: 6pt, stroke: 0.75pt +
rgb(226, 232, 240), ) [ align(center) [ text(size: 11pt, weight: "bold", fill: rgb(51, 65, 85)) [小组成员]
v(0.55cm) grid( columns: (1fr, 1fr), column-gutter: 1.6cm, row-gutter: 0.65cm, align: center +
horizon, [text(size: 13pt, weight: "bold") [张恒基 (组长) ]
text(size: 12pt, fill: rgb(100, 116, 139)) [2024211301 / 2024210926]], [text(size: 13pt, weight: "bold")
[秦博宇]
text(size: 12pt, fill: rgb(100, 116, 139)) [2024211302 / 2024210940]], [text(size: 13pt, weight: "bold")
[陈艺博]
text(size: 12pt, fill: rgb(100, 116, 139)) [2024211302 / 2024210931]], [text(size: 13pt, weight: "bold")
[陈雨飞]
text(size: 12pt, fill: rgb(100, 116, 139)) [2024211301 / 2024210918]], ) ] ]
```

```
v(1cm) text(size: 12pt, fill: rgb(100, 116, 139)) [工程交付 · 2026-06-18] v(0.5cm) text(size: 10.5pt,
fill: rgb(148, 163, 184)) [北京邮电大学 · 计算机学院] ]
```

环境要求

- **JDK 21:** java -version 确认
- **Maven 3.9+:** mvn -version 确认
- Windows / macOS / Linux 均可

构建

`mvn clean package -DskipTests`

运行模式

模式	命令	预期行为
交互菜单	<code>mvn exec:java -f jieqi-app/pom.xml -am</code>	显示 1-9 菜单
WS 服务器	<code>mvn exec:java -f jieqi-app/pom.xml -am -Dexec.args="server-ws 8887"</code>	监听 8887 端口
WS 客户端	<code>mvn exec:java -f jieqi-app/pom.xml -am -Dexec.args="client-ws ws://127.0.0.1:8887 player1 123456"</code>	连接并登录
AI 自动对弈	<code>mvn exec:java -f jieqi-app/pom.xml -am -Dexec.args="ai-ws ws://127.0.0.1:8887 ai_bot_1 pw123"</code>	自动走子
TCP 服务器	<code>mvn exec:java -f jieqi-app/pom.xml -am -Dexec.args="server 8888"</code>	监听 8888
TCP 客户端	<code>mvn exec:java -f jieqi-app/pom.xml -am -Dexec.args="client 127.0.0.1 8888 player1 123456"</code>	连接
Fat JAR	<code>java -jar jieqi-app/target/unveil-jieqi.jar</code>	单 JAR 启动

完整演示流程

终端 1: `mvn exec:java -f jieqi-app/pom.xml -am -Dexec.args="server-ws 8887"`

终端 2: `mvn exec:java -f jieqi-app/pom.xml -am -Dexec.args="client-ws ws://127.0.0.1:8887 player1 123456"`

```
> login
> match
> ready
> first true
```

终端 3: `mvn exec:java -f jieqi-app/pom.xml -am -Dexec.args="client-ws ws://127.0.0.1:8887 player2 123456"`

```
> login
> match
> ready
> first false
```

对局中

```
> move b 0 b 3
> ai medium
> replay
> rematch
```

自检

`powershell -File scripts/verify.ps1`

Docker 部署

```
docker compose up --build
```

```
# 端口 8887 (WS)、8888 (TCP) 映射
```